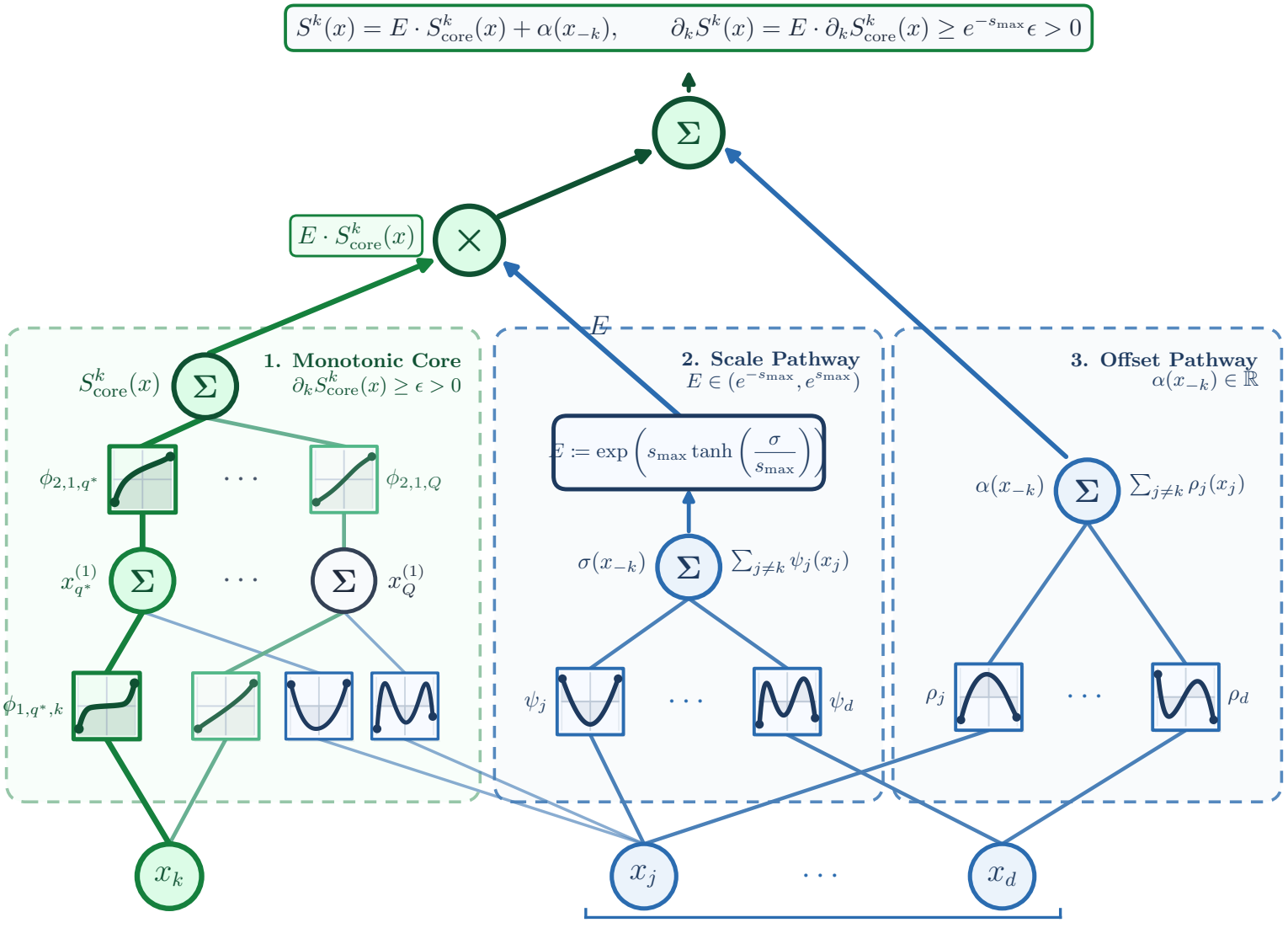

$S^k(x) = E \cdot S^k_{\text{core}}(x) + \alpha(x_{-k}), \qquad \partial_k S^k(x) = E \cdot \partial_k S^k_{\text{core}}(x) \geq e^{-s_{\max}}\epsilon > 0$
$E \cdot S^k_{\text{core}}(x)$
E
1. Monotonic Core
$\partial_k S^k_{\text{core}}(x) \geq \epsilon > 0$
$S^k_{\text{core}}(x)$
$\phi_{2,1,q^*}$
$\phi_{2,1,Q}$
$x^{(1)}_{q^*}$
$x^{(1)}_Q$
$\phi_{1,q^*,k}$
x_k
2. Scale Pathway
$E \in (e^{-s_{\max}}, e^{s_{\max}})$
$E := \exp\left(s_{\max} \tanh\left(\frac{\sigma}{s_{\max}}\right)\right)$
$\sigma(x_{-k})$
$\sum_{j\neq k} \psi_j(x_j)$
ψ_j
ψ_d
3. Offset Pathway
$\alpha(x_{-k}) \in \mathbb{R}$
$\alpha(x_{-k})$
$\sum_{j\neq k} \rho_j(x_j)$
ρ_j
ρ_d
x_j
x_d
Target x_k (enters S^k_{core} only)
Conditioning $x_{-k} = (x_j)_{j\neq k}$ (unconstrained across S^k_{core}, σ, and α)